

PERCENTAGES, PROFIT AND LOSS

Q1

Directions for question 1: Type in your answer in the input box provided below the question.

What is the value of $8\frac{1}{3}\%$ of 600 + $37\frac{1}{2}\%$ of 400?

A man bought a clock at $\frac{3}{4}$ th of its list price and sold it at 50% above its list price. What is his profit percentage in the transaction?

- ☐ a) 50%
- ☐ b) 70%
- ☐ c) 80%
- ☐ d) None of these

Solution:

If the list price (or marked price) is 100, the cost price is 75 and the selling price is 150. Therefore, there is a profit of 100% in the transaction. Choice (D)

A and B invested Rs.30,000 and Rs.40,000 respectively in a business for one year. If they together earned Rs.7,700 at the end of one year, then find B's profit.

- ☐ a) Rs.3,800
- ☐ b) Rs.3,300
- ☐ c) Rs.4,400
- ☐ d) Rs.4,000

(Correct Answer - ✓)

Solution:

The amounts invested by A and B are in the ratio 3:4. Therefore, B's share in the profit is $\frac{4}{7}$ (7700), i.e. ₹ 4400. Choice (C)

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

Find the annual income of a person who invested Rs.11000 in a 5% stock at 10% premium.

Step 1: Understand “5% stock”

- A **5% stock** means: for every **₹100 face value**, the annual income (dividend) is **₹5**.

Step 2: Account for premium

- Stock is bought at **10% premium**, i.e. **₹110 per ₹100 face value**. So, to invest ₹11,000, the face value purchased is:

$$\text{Face Value} = 11,000 \div 110 \times 100 = 10,000$$

Step 3: Calculate annual income

- Dividend = **5% of face value**

$$\text{Income} = 5\% \times 10,000 = 500$$

☒ Final Answer

The person's **annual income** = **₹500**.

Directions for questions 4 and 5: Type in your answer in the input box provided below the question.

The production of rice went up from 2000 million tonnes in 1998 to 2300 million tonnes in 1999. Find the percentage increase in the production of rice from 1998 to 1999.

Your Answer: Unattempted

Correct Answer: 15 ✓

Directions for questions 6 and 7: Select the correct alternative from the given choices.

P sold an article for Rs.1,080 thereby losing 10%. Q sold another article for Rs.1,800 at a loss of 10%. Who incurred a greater loss?

☐ a) P

☒ b) Q (Correct Answer - ✓)

☐ c) Cannot say

☐ d) Both have an equal loss

Solution:

P sold the article at a loss of 10% .

$$\therefore 0.9 CP_1 = 1080 \Rightarrow CP_1 = 1200.$$

$\therefore$ P lost ₹120

Q sold the circle at a loss of 10%

$$\Rightarrow CP_2 = \frac{1800}{0.9} = 2000$$

$\therefore$ Q lost ₹200.

Choice (B)

If Raju's salary went up by 40%, by what percentage should it be brought down to bring it back to the original level?

☐ a) $19\frac{1}{3}\%$

☐ b) $16\frac{2}{3}\%$

☒ c) $28\frac{4}{7}\%$

(Correct Answer - ✓)

☐ d) $23\frac{1}{7}\%$

Solution:

If a quantity increases from 100, by 40%, it becomes 140. It has to now decrease by $\frac{40}{140}$ (100%), i.e. $\frac{200}{7}\%$ or $28\frac{4}{7}\%$ to bring it back to the original level. Choice (C)

Directions for questions 8 and 9: Type in your answer in the input box provided below the question.

A man sold an article at 10% gain. If he had charged Rs.45 more his gain would have been 25%. What is the cost price (in Rs.)?

Solution:

The man sold the article at a profit of 10%. If he had charged ₹ 45 more, the profit would have been 25%, i.e. 15% of C (the cost price) more. 15% of C is 45, 100% of C is ₹300. Ans : (300)

Directions for questions 8 and 9: Type in your answer in the input box provided below the question.

Two numbers are less than a third number by 40% and 46% respectively. By what percent is the second number less than the first?

Solution:

If the third number is 100, the first is 60 and the second is 54. The second is less than the first by $\frac{6}{60}$ (100%), i.e. 10%.
Ans : (10)

Directions for questions 10 and 11: Select the correct alternative from the given choices.

The length of a rectangle is increased by 30% where as the breadth is decreased by 25%. What is the consequent percentage change in the area of the rectangle?

- ☐ a) 2.5% increase
- ☐ b) 25% decrease
- ☐ c) 5% increase
- ☐ d) 2.5% decrease

Solution:

Let the original length and breadth be l and b respectively. The new length and breadth are $1.3l$ and $0.75b$. The original area is lb and the new area is $\frac{13}{10} \times \frac{3}{4} lb = \frac{39}{40} lb$. The fractional decrease is $\frac{1}{40}$ and the percentage decrease is 2.5%
Choice (D)

Directions for questions 10 and 11: Select the correct alternative from the given choices.

Sankar, Srinivas and Manohar start a business with Rs.40,000, Rs.30,000 and Rs.70,000 respectively. Srinivas and Manohar leave after 5 months and 7 months respectively. Who gets the largest share of profit at the end of the year?

- ☐ a) Sankar
- ☐ b) Srinivas
- ☐ c) Manohar
- ☐ d) Cannot be determined

Solution:

The data is tabulated below.

	Sankar	Srinivas	Manohar
Amount	40,000	30,000	70,000
Period	12	5	7
Profit shares	4(12)	3(5)	7(7)

Manohars share is the largest.

Choice (C)

Directions for question 12: Type in your answer in the input box provided below the question.

The salary of all officers is increased twice successively by 20%. What is the net percentage increase in their salaries?

Solution:

The salary increases from (say) 100 to 120 and then to 144. There is a net increase of 44%

Ans : (44)

Directions for questions 13 and 14: Select the correct alternative from the given choices.

Arun purchased a house for Rs.75,000 and a site for Rs.15,000 respectively, if he sold the house for Rs.83,000 and the site for Rs.10,000, then find the resultant percentage of gain.

- ☐ a) 3%
- ☐ b) $3\frac{1}{3}\%$
- ☐ c) 30%
- ☐ d) $33\frac{1}{3}\%$

Solution:

The total cost incurred on the two purchases is 90,000. The total amount realised on the two sales is 93,000. There is a net profit of $\frac{3}{90}$ (100%), i.e. $3\frac{1}{3}\%$.

Choice (B)

Two numbers are respectively 30% and 40% of a third number. What percent is the first of the second?

- ☐ a) 25%
- ☐ b) 50%
- ☐ c) 40%
- ☐ d) 75%

Solution:

If the third number is 100, the first is 30 and the second is 40. The first number is $\frac{30}{40} (100) \%$, i.e. 75 % of the second. Choice (D)

Directions for question 15: Type in your answer in the input box provided below the question.

Srinath and Prasad invested Rs.36,000 and Rs.40,000 respectively to start a firm. Prasad withdrew after nine months. If they earned a profit of Rs.55,000 at the end of the year, then find Srinath's share (in Rs.).

Solution:

The data and the simplification is tabulated below.

	Srinath	Prasad
Amount	36,000	40,000
Period (month)	12	9
Profit shares	9(12)	10(9)
Profit shares	6	5

Srinath's share is $\frac{6}{11}$ (55000), which is 30,000.

Ans : (30000)